LIGHTYEAR

MS82 Binary overlay and variable record decoding

Customer benefit

Prepare to read the bank record formats that go beyond CardDemo's fixed DISPLAY layouts, before customer runtime access arrives. Exact byte interpretation stays separate from business equivalence.

Delivered behavior

The copybook compiler supports IBM COMP/BINARY/COMP-4 and COMP-5 fields with exact implied decimal scaling, REDEFINES storage overlays, and one bounded OCCURS DEPENDING ON table with sliding trailing fields. Binary byte order and truncation profile are explicit. Every overlay requires a selected alternative.

Record framing accepts a fixed stream, one payload, or unblocked four-byte RDW records. Counts, lengths, bounds and reserved RDW bytes are checked. Unknown choices, truncation, nested ODO, ODO inside repeated/overlaid areas, alignment, BDWs and spanned records fail instead of guessing.

Evidence and limits

Extended decoder guide, copybook, and independent synthetic byte vectors.

Original bytes, padding, copybook/record hashes and exact values are retained. Existing CardDemo fixed-record evidence and the 501-record public decoding receipt remain unchanged. No native customer run, compiler qualification or business-equivalence rule is claimed.

Validation

Independent vectors cover signed/unsigned 2/4/8-byte fields, endian selection, 64-bit extremes, implied decimals, different overlay sizes, per-occurrence overlay choices, zero/minimum/maximum counts, sliding fields and malformed RDWs. CLI tests exercise explicit profiles and failure without success output. Existing public CardDemo and loopback batch/spool tests remain part of CI.